Areg Hovumyan

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Education

California State Polytechnic University, Pomona — Pomona, CA
Bachelor of Science in Computer Engineering — GPA: 3.6/4.0 — (rising senior) Expected December 2027

Experience and Projects

Electrical/Embedded Systems Member: L'SPACE (Mission Concept) — Remote
NASA's student acceleration program - Robotic Lunar Rover Mission — May 2026 – Present

- Present the Communication and Data Handling to NASA engineers for a possible mission consideration on the moon.
- Manage to stay within the schedule, cost, and mass limitations (under 250M, 170Kg, no sunlight for 14 days and more).
- Design the whole rover's Communication and Data Handling(C&DH) architecture by calculating the best fitting battery, figuring out the communication protocols, and choosing the flight software by studying from NASA's previous missions.
- Establish (on concept) the communication between the Earth and the Moon, autonomous mission completion in case of redundancy, and learn F' to fit the mission under the mission needs.

Software Engineering Team Lead: Autonomous Hexacopter (University Club) — Pomona, CA
Lockheed Martin sponsored project — June 2025 – Present

- Apply pair programming, test-first development, unit-testing, create containers for a ROS2 environment with Docker for a team of 8 student-engineers to collaborate, and created CI/CD pipelines to ship features with fewer defects.
- Collaborate with the Electrical team to not overwhelm the drone with object detection and image capturing, and work with the Payload team to drop payloads mid-flight using servos.
- Present mapping and Computer vision improvement, image processing achievement, and correct GPU/CPU utilization to Lockheed Martin engineers during the Preliminary Design Review (PDR).
- Create ROS2 nodes handling the whole mission, including object detection (0.99 mAP@0.5), area mapping, 4k image processing (under 0.7s each), and payload release over a target.

Embedded Systems Member: Autonomous Ground Vehicle (University Club) — Pomona, CA
Northrop Grumman Collaboration Project — February 2026 – June 2026

- Present telemetry connection between a drone, rover, and a ground station workflow to Northrop Grumman engineers during a Preliminary Design Review (PDR).
- Implement Feature-driven development and the lean approach to release new code every week and meet all the requirements before any demonstrations or meetings.
- Run an authorized hardware review of a DS2431-class authenticator with a Bus Pirate v4 and logic analyzer, confirming a family-code mismatch and identifying clone silicon.
- Collaborate with the firmware team to develop bare-metal firmware for a pre-made microcontroller in C, directly manipulating registers (GPIO, timers, interrupts, etc.).

Software Engineer: Autonomous Vehicle Lab (Research) — Pomona, CA
University Research Lab — April 2026 – Present

- Creating an interceptor drone, running object detection on a Jetson, establishing autonomous flight through a ready-made flight controller, and fitting under the budget of $2000.
- Build a closed-loop stepper controller (CL57T) on a microcontroller with CAN telemetry over a TJA1051 transceiver to control a steering wheel through a computer, controller, or a joystick.
- Engineered CI/CD pipeline in GitHub Actions for the team to catch regressions before merge, and create unit tests to improve system reliability.
- Resolve priority-inversion bug by introducing mutex priority inheritance, restoring deterministic timing.
- Utilize AI agents to interface an MPU IMU over I2C on a Nucleo to stream 9-DOF telemetry.

Technical Skills and Relevant Coursework

Languages and Tools: C, C++, Python, Rust, Java, VHDL, ARM Assembly, Bash, SQL, bare-metal, UART, SPI, ROS2 (Humble/Iron), Gazebo, YOLO, PyTorch, computer vision, machine learning, Linux, Git, GitHub Actions, Docker, MongoDB.
Coursework: Microcontrollers & Embedded Systems, Circuit Analysis, Signals & Systems, Data Structures & Algorithms.
Languages: English, Russian, Armenian